Python Code

```python

"""

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score
from sklearn.feature_selection import SequentialFeatureSelector
from sklearn.linear_model import LinearRegression


data = {
    'Rainfall_mm': [1200, 450, 2100, 800, 1500, 600, 1800, 500, 1300, 1000],
    'Temp_C': [24.5, np.nan, 21.0, 27.5, 19.5, 29.0, 22.5, 35.0, 25.0, 26.0],
    'Soil_Moisture': [35, 12, 55, 25, 40, 18, 48, 15, 38, 30],
    'Soil_pH': [6.5, 7.8, 5.2, 7.0, 6.0, 7.5, 5.8, 8.2, 6.4, 6.9],
    'Nitrogen': [80, 30, 110, 55, 95, 45, 105, 35, 85, 70],
    'Phosphorus': [45, 15, 60, 30, np.nan, 25, 55, 20, 48, 38],
    'Potassium': [210, 120, 280, 180, 240, 150, 265, 130, 220, 195],
    'Irrigation': [85, 20, 95, 50, 75, 35, 90, 15, 80, 65],
    'Crop_Area': [4500, 1200, 8000, np.nan, 5500, 2200, 6800, 1500, 4800, 4000],
    'Fertilizer': [180, 60, 250, 120, 210, 90, 230, 70, 190, 150],
    'Pesticide': [4.2, 1.5, 8.0, 3.1, 5.5, 2.4, 7.2, 1.8, 4.8, 3.8],
    'Sunlight': [8.5, 10.0, 5.5, 7.0, 6.5, 9.0, 6.0, 11.0, 8.0, 7.5],
    'Humidity': [65, 35, 85, 50, 75, 40, 80, 30, 70, 60],
    'Altitude': [150, 50, 1200, 400, 1800, 150, 900, 100, 250, 600],
    'Wind_Speed': [12, 25, 10, 18, 8, 30, 12, 28, 14, 16],
    'Groundwater': [15, 45, 8, 22, 5, 38, 10, 50, 18, 28],
    'Diversity_Index': [0.75, 0.30, 0.85, 0.55, 0.80, 0.45, 0.82, np.nan, 0.70, 0.65],
    'Mechanization': [80, 25, 90, 60, 75, 40, 85, np.nan, 82, 70],
    'Yield_Variance': [0.12, 0.25, 0.08, 0.18, 0.10, 0.20, 0.09, np.nan, 0.13, 0.15],
    'Avg_Yield': [8.2, 2.1, 12.5, 5.4, 9.8, 3.8, 11.2, 2.5, 8.8, 6.7]
}

df = pd.DataFrame(data)
df

"""EDA ANALYSIS

"""

#BASIC DATA INFORMATION
print(df.head())
print(df.tail())
print(df.info())

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#SHAPE AND SIZE
print(df.shape)
print(df.size)

#NULL VALUES
print(df.isna().sum())
# Fill null values with the mean of their respective columns
df = df.fillna(df.mean())

# Verify that there are no more null values
print(df.isnull().sum().sum())

"""CORELATION ACROSS REGIONS"""



df.corr(numeric_only=True)
sns.heatmap(df.corr(numeric_only=True))
plt.title("Correlation Heatmap")
plt.show()

"""DATA NORMALIZATION [MIN-MAX SCALER]"""

scaler = StandardScaler()
scaled_array = scaler.fit_transform(df)
df_normalized = pd.DataFrame(scaled_array, columns=df.columns)
print(df_normalized)

import seaborn as sns
sns.boxplot(df)
plt.show()

"""FORWARD/BACKWARD FEATURE SELECTION"""

x = df.iloc[:,:-1]
y = df.iloc[:,-1]
model = LinearRegression()
fs = SequentialFeatureSelector(model, n_features_to_select=3, direction='backward')
fs.fit(x, y)

"""K-CLUSTERING"""

X = df.values
wcss = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel("Number of Clusters (k)")
plt.ylabel("WCSS")
plt.title("Elbow Method")
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plt.show()

# before clustering
plt.figure()
#to plot 'Rainfall_mm'
plt.scatter(df['Rainfall_mm'], [0]*len(df))
plt.title("Before Clustering - Rainfall_mm Distribution")
plt.xlabel("Rainfall_mm Value")
plt.yticks([])
plt.show()

# After clustering
plt.figure()

optimal_k = 3
kmeans_final = KMeans(n_clusters=optimal_k, random_state=42, n_init='auto')
kmeans_final.fit(X)
df['Cluster'] = kmeans_final.labels_
#cluster centroids
centroids = kmeans_final.cluster_centers_
plt.scatter(df["Rainfall_mm"], [0]*len(df), c=df["Cluster"])
plt.scatter(centroids[:, 0], [0]*len(centroids), marker='h', s=200, color='red', edgecolors='black')

plt.title(f"After Clustering (k={optimal_k}) - Rainfall_mm Distribution")
plt.xlabel("Rainfall_mm Value")
plt.yticks([])
plt.show()

print("\nClustered Data:")
print(df)
wcss

"""SILHOUETTE SCORE"""

silhouette_avg = silhouette_score(X, df['Cluster'])
print(f"The average silhouette score is: {silhouette_avg}")

""" PCA FOR 2D VISUALIZATION"""

pca = PCA(n_components=2)
pca_data = pca.fit_transform(X)

plt.figure(figsize=(10, 6))
sns.scatterplot(x=pca_data[:, 0], y=pca_data[:, 1], hue=df['Cluster'], palette='viridis', s=200)
plt.title('Agricultural Regions Clustered (PCA View)')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.show()

"""MEAN CHARACTERISTIC OF EACH CLUSTER"""

print("\n--- Cluster Profiles (Mean Values) ---")
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print(df.groupby('Cluster').mean(numeric_only=True))

"""PREDICTED CLUSTER"""

predicted_clusters = kmeans_final.predict(X)
df['Predicted_Cluster_Label'] = predicted_clusters
print("DataFrame with Predicted Cluster Labels:")
print(df[['Rainfall_mm', 'Avg_Yield', 'Cluster', 'Predicted_Cluster_Label']].head())

"""PREDICTION FOR NEW DATA"""


```